

Generative AI Project Proposal

Visible-to-Infrared Image Translation via GANs

Sahrish Mustafa (22i0977)
Aaliyan Raza Kazmi (22i0833)
Aashir Hameed (22i0920)

Department of Computer Science
FAST – National University of Computer and Emerging Sciences
October 2025

Contents

1	Introduction and Motivation	2
2	Related Work	3
2.1	GAN-based Approaches	3
2.2	Diffusion-based Approaches	4
2.3	Transformer-based Approaches	4
2.4	Summary	4
3	Proposed Methodology	5
3.1	Model Architecture	5
3.2	Loss Functions and Training	5
3.3	Data and Infrared Band Selection	6
3.4	Note on Scope and Flexibility	6
	Note on Scope and Flexibility	6

CHAPTER 1

Introduction and Motivation

Infrared (IR) imagery plays a crucial role in applications where visible-light sensing is insufficient, such as nighttime surveillance, autonomous driving, and environmental monitoring. Unlike standard RGB cameras, long-wave infrared (LWIR) sensors (typically 7.5–14 micrometers) capture emitted thermal radiation, allowing detection of objects independent of illumination. For example, LWIR cameras can reveal pedestrians or animals at night that RGB cameras miss, and are not affected by headlight glare or fog. However, collecting large-scale, high-quality IR datasets is costly. Visible-to-infrared (V2IR) image translation thus aims to synthesize realistic IR images from abundant RGB data, enabling data augmentation and robust model training for downstream tasks without expensive sensors [Lynred \[2023\]](#).

V2IR translation is challenging because it must map color images to thermal signatures under physical laws. Visible scenes have ambiguous thermal interpretations, and IR cameras operate in varied wavelength bands (SWIR, MWIR, LWIR) with different characteristics. The key challenges include achieving semantic-aware mapping, handling diverse wavelength bands, and overcoming limited paired RGB–IR data. GANs have shown promise for image-to-image translation, but typical VIS→IR models often produce artifacts or blurred details. The goal of this project is to develop a GAN-based V2IR model inspired by InfraGAN [Özkanoglu and Özer \[2022\]](#) that synthesizes high-quality thermal images from RGB inputs.

CHAPTER 2

Related Work

2.1 GAN-based Approaches

Generative Adversarial Networks (GANs) have been the foundation of most early image-to-image translation tasks. **Pix2Pix** [Isola et al. \[2017\]](#) introduced the conditional GAN (cGAN) framework for paired translation, where a U-Net generator and a PatchGAN discriminator jointly learn a mapping from RGB to target domains using both adversarial and L1 reconstruction losses. **CycleGAN** [Zhu et al. \[2017\]](#) extended this to unpaired datasets through cycle-consistency loss, enforcing $G(F(x)) \approx x$. However, both frameworks can suffer from mode collapse and often yield deterministic or overly smooth IR outputs.

Later models such as **MUNIT** and **DRIT** introduced content–style disentanglement, allowing multiple outputs per input. Although powerful for stylistic transfer, these frameworks were not originally designed for physically constrained thermal imagery.

In the V2IR domain, **ThermalGAN** [Kniaz and et al. \[2018\]](#) and **InfraGAN** [Özkanoglu and Özer \[2022\]](#) improved IR generation quality using cGAN backbones with structural losses (e.g., SSIM) to enhance realism and edge preservation. Further enhancements like **EGGAN** and **DR-AVIT** [Han et al. \[2024\]](#) added edge and geometry consistency to improve cross-domain alignment.

Recently, more advanced GAN variants have been proposed to improve domain fidelity and semantic consistency in thermal generation. **IRKDGAN (Discriminator-Guided GAN)** [Chen et al. \[2023\]](#) introduces a dual-purpose discriminator that not only classifies real and fake images but also reconstructs infrared features, acting as a “teacher” for the generator via knowledge distillation. This design allows the generator to learn more robust thermal representations and improves structural accuracy in generated IR images. **TM-GAN (Template-Matching GAN)** [Zhang et al. \[2024\]](#) incorporates a template-matching module into the adversarial framework by introducing a normalized cross-correlation (NCC) and match-patch loss. These losses explicitly align synthesized

IR patterns with real IR templates, significantly improving edge correspondence and downstream object detection performance in thermal imagery.

2.2 Diffusion-based Approaches

Diffusion-based models have recently achieved state-of-the-art results in thermal image generation. **PID (Physics-Informed Diffusion)** [Mao et al. \[2024\]](#) integrates thermal physics by decomposing IR images into Temperature, Emissivity, and Texture (TeV) components, ensuring physical correctness through dedicated losses. **DiffV2IR** [Ran et al. \[2025\]](#) further extends this by combining large-scale IR pretraining and vision-language conditioning for improved semantic awareness under data-scarce conditions.

2.3 Transformer-based Approaches

Transformer-based architectures like **IRFormer** [Liu et al. \[2024\]](#) utilize self-attention mechanisms to capture global relationships and mitigate common GAN limitations, such as misinterpreting luminance-correlated features. These models maintain strong global coherence while enhancing fine local detail, marking a shift toward attention-driven generative models for thermal imaging.

2.4 Summary

V2IR research has evolved from classical GANs (*Pix2Pix*, *CycleGAN*) to physics-informed and attention-driven approaches (*InfraGAN*, *PID*, *DiffV2IR*, *IRFormer*). Our proposed model builds upon these insights by combining the structural strengths of cGANs with SSIM and perceptual losses to generate thermally consistent and semantically aligned LWIR imagery.

CHAPTER 3

Proposed Methodology

3.1 Model Architecture

We propose a **conditional Generative Adversarial Network (cGAN)** inspired by Pix2Pix [Isola et al. \[2017\]](#). The generator employs a **U-Net** encoder–decoder with skip connections to preserve fine spatial details, converting a 3-channel RGB input to a 1-channel IR output. The discriminator uses a **PatchGAN** design, classifying overlapping patches instead of entire images to enforce local realism. Following **InfraGAN** [Özkanoglu and Özer \[2022\]](#), an SSIM-based structural loss is integrated to enhance edge fidelity and preserve physical luminance consistency.

3.2 Loss Functions and Training

To balance realism and structure preservation, multiple loss terms are used:

- **Adversarial Loss** (L_{adv}) – Matches generated IR distribution to real IR samples using Least Squares GAN.
- **Reconstruction Loss** (L_1) – Ensures overall brightness and intensity alignment with paired data.
- **SSIM Loss** ($1 - \text{SSIM}$) – Encourages local structural similarity and edge preservation.
- **Perceptual Loss** (L_{perc}) – Enhances visual realism using pretrained VGG feature space.
- **Edge Loss** (L_{edge}) – Optional edge-based penalty for sharper contours.

The combined generator objective is:

$$L_G = \lambda_{adv}L_{adv} + \lambda_{L1}L_1 + \lambda_{SSIM}(1 - \text{SSIM}) + \lambda_{perc}L_{perc} + \lambda_{edge}L_{edge}.$$

3.3 Data and Infrared Band Selection

Training will utilize the **KAIST Multispectral** and **FLIR ADAS** datasets [Mao et al. \[2024\]](#), which provide aligned RGB–LWIR image pairs. We focus on the **long-wave infrared (LWIR)** range (7.5–13.5 μm), the most practical for automotive and surveillance applications. This allows the model to capture physically realistic heat emissions, where warmer objects (humans, vehicles) appear brighter than cooler surfaces (roads, background).

3.4 Note on Scope and Flexibility

It is important to note that the architectural choices, training procedures, and evaluation metrics outlined in this proposal are preliminary and may evolve as the project progresses. The proposed design serves as a conceptual foundation; specific hyperparameters, implementation details, or loss formulations may be adjusted based on empirical results, computational feasibility, or dataset constraints.

Bibliography

Hao Chen, Yuxuan Liu, Xinyu Wang, and Ming Zhou. Infrared image translation via discriminator-guided gan with knowledge distillation. *Multimedia Tools and Applications*, 82(11):17023–17039, 2023. doi: 10.1007/s11042-023-14618-9. URL <https://link.springer.com/article/10.1007/s11042-023-14618-9>.

J. Han et al. Dr-avit: Towards diverse and realistic aerial visible-to-infrared image translation. *ResearchGate*, 2024. URL https://www.researchgate.net/publication/380918863_DR-AVIT_Towards_Diverse_and_Realistic_Aerial_Visible-to-Infrared_Image_Translation.

Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. URL https://openaccess.thecvf.com/content_cvpr_2017/papers/Isola_Image-To-Image_Translation_With_CVPR_2017_paper.pdf.

Vladimir Kniaz and et al. Thermalgan: Multimodal color-to-thermal image translation for person re-identification in multispectral dataset. In *Proceedings of ECCV Workshops*, 2018. URL https://openaccess.thecvf.com/content_ECCVW_2018/papers/11134/Kniaz_ThermalGAN_Multimodal_Color-to-Thermal_Image_Translation_for_Person_Re-Identification_in_Multispectral_ECCVW_2018_paper.pdf.

Jun Liu, Jin Xu, Zhaoli Zheng, and Baoxin Sun. Implicit multi-spectral transformer: An lightweight and effective visible to infrared image translation model. *arXiv preprint arXiv:2404.07072*, 2024. URL <https://arxiv.org/abs/2404.07072>.

Lynred. How thermal imaging is contributing to the development of a new generation of nighttime pedestrian detection systems. <https://www.lynnred.com/blog/how-thermal-imaging-contributing-development-new-generation-nighttime-pedestrian-detection> 2023.

X. Mao et al. Pid: Physics-informed diffusion model for infrared image generation. *arXiv preprint arXiv:2407.09299*, 2024. URL <https://arxiv.labs.arxiv.org/html/2407.09299v1>.

- L. Ran et al. Diffv2ir: Visible-to-infrared diffusion model via vision-language understanding. *arXiv preprint arXiv:2503.19012*, 2025. URL <https://arxiv.org/abs/2503.19012>.
- Wei Zhang, Zhe Li, Chengjie Wang, and Rui Huang. Tm-gan: Template-matching generative adversarial network for visible-to-infrared image translation. *arXiv preprint arXiv:2403.11245*, 2024. URL <https://arxiv.org/abs/2403.11245>.
- Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017. URL https://openaccess.thecvf.com/content_ICCV_2017/papers/Zhu_Unpaired_Image-To-Image_Translation_ICCV_2017_paper.pdf.
- A. Özkanoglu and M. Özer. Infragan: A gan architecture to transfer visible images to infrared domain. *OpenReview*, 2022. URL <https://openreview.net/forum?id=dUIktA2SrV>.